

# MULTI-AGENT SERVING: WORKLOADS → OPTIMIZATION FAMILIES

| Workload characteristic                                    | Memory optimization                        | Prefix-aware scheduling                            | Cache lifecycle                                 | Power management                   |
|------------------------------------------------------------|--------------------------------------------|----------------------------------------------------|-------------------------------------------------|------------------------------------|
| <b>Sparse activation</b><br>few agents generate at once    | <b>ScaleSim</b><br>prefetch / eviction     | Activity-aware placement                           | Idle KV reclamation                             | <b>KAIROS</b><br>phase-aware power |
| <b>Invocation distance</b><br>predictable next-use gap     | <b>ScaleSim</b><br>distance-aware eviction | <b>TOPAS / TokenCake</b><br>future-aware placement | <b>Continuum</b><br>TTL / resume policy         | Idle phases expose slack           |
| <b>Inter-agent sharing</b><br>shared prompts / KV prefixes | Shared-state memory pressure               | <b>TokenCake / KVFlow</b><br>prefix-aware routing  | <b>ForkKV / KVCOMM</b><br>reuse / copy-on-write | Indirect efficiency effect         |

Current gap: surveyed systems optimize subsets of the stack; diagnostic signals are not a first-class serving input.